D.A.A Lab Assignment 9:

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Q1 - Floyd Warshall Algo Code:

main.cpp

```cpp
#include <iostream>
using namespace std;

#define INF 99999
#define MAX 100

int main()
{
    int n, graph[MAX][MAX], dist[MAX][MAX];

    cout << "Enter the number of vertices: ";
    cin >> n;

    cout << "Enter the adjacency matrix (use 99999 for INF):\n";

    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < n; j++)
        {
            cin >> graph[i][j];
            dist[i][j] = graph[i][j];
        }
    }

    for (int k = 0; k < n; k++)
    {
        for (int i = 0; i < n; i++)
        {
            for (int j = 0; j < n; j++)
            {
                if (dist[i][k] != INF && dist[k][j] != INF &&
                    dist[i][k] + dist[k][j] < dist[i][j])
                {
                    dist[i][j] = dist[i][k] + dist[k][j];
                }
            }
        }
    }

```

```
39
40      cout << "\nShortest distance matrix:\n";
41
42      for (int i = 0; i < n; i++)
43      {
44          for (int j = 0; j < n; j++)
45          {
46              if (dist[i][j] == INF)
47                  cout << "INF\t";
48              else
49                  cout << dist[i][j] << "\t";
50          }
51          cout << endl;
52      }
53
54      return 0;
55  }
```

Output:

```
Enter the number of vertices: 3
Enter the adjacency matrix (use 99999 for INF):
0
2
4
2
0
1
4
1
0

Shortest distance matrix:
0       2       3
2       0       1
3       1       0


...Program finished with exit code 0
Press ENTER to exit console.
```